

Database Development

Database development is the process of designing, creating, implementing, maintaining, and optimizing databases that store and manage application data efficiently, securely, and reliably.

Objectives of Database Development

Store data systematically, ensure integrity, improve performance, maintain security, and support scalability.

Database Architecture

Application Layer -> Data Access Layer -> DBMS -> Physical Storage

Database Development Lifecycle

Requirements Analysis -> Design -> Schema Creation -> Implementation -> Testing -> Deployment -> Maintenance

Database Design

Conceptual design identifies entities and relationships among data objects.

Database Models

Relational databases (MySQL, PostgreSQL, SQL Server) and NoSQL databases (MongoDB, Cassandra, Redis).

Database Schema Design

Tables, keys, constraints, and relationships define data structures.

Database Normalization

1NF, 2NF, and 3NF reduce redundancy and improve consistency.

CRUD Operations

Create, Read, Update, and Delete operations manage records.

Database Security

Authentication, RBAC, encryption, backups, audit logging, and SQL injection prevention.

Performance Optimization

Indexing, query optimization, partitioning, caching, and connection pooling.

Backup and Recovery

Full, Incremental, and Differential backups with recovery procedures.

Database Testing

Schema validation, integrity, performance, security, and recovery testing.

Deployment Architecture

Application Server -> Primary Database -> Read Replica / Backup Server

Benefits of Database Development

- Efficient data management
- High data integrity
- Improved security
- Better performance
- Scalability
- Reliable backup and recovery
- Support for business analytics